Marwadi University	Marwari University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing(01CT0513)	**Aim: OPEN HANDED** Apply non-linear filters on images and investigate its application in noise-removal.	
Experiment No: 09	**Date: 01-10-2025**	**Enrollment No:92301733054**

Aim: Apply non-linear filters on images and investigate its application in noise-removal.

Theory:-

- Non-linear filters are image processing techniques used for noise removal by considering the local neighborhood of each pixel. Unlike linear filters, non-linear filters modify pixel values based on their relationship with neighboring pixels, allowing them to effectively suppress different types of noise.

Programm:-

```
import cv2
from google.colab.patches import cv2_imshow
import numpy as np
import os
image_files = [

"/content/smoothening data.png",
"/content/smoothening data2.png",
"/content/smoothening data3.png",
"/content/smoothening data4.png",
"/content/smoothening data5.png",
]
output_dir = "/content/filtered_images"
os.makedirs(output_dir, exist_ok=True)
filtered_images = []
for idx, file in enumerate(image_files, start=1):
    img = cv2.imread(file)

if img is None:

print(f"Could not read {file}")

continue
    filtered_img = cv2.GaussianBlur(img, (7, 7), 0)
    filtered_images.append(filtered_img)
    save_path = os.path.join(output_dir, f"filtered_image_{idx}.png")
    cv2.imwrite(save_path, filtered_img)

print(f"Saved: {save_path}")
Saved: /content/filtered_images/filtered_image_5.png
heights = [img.shape[0] for img in filtered_images]
widths = [img.shape[1] for img in filtered_images]
min_height = min(heights)
min_width = min(widths)
resized_images = [cv2.resize(img, (min_width, int(img.shape[0] * min_width / img.shape[1])))

for img in filtered_images]
grid = np.vstack(resized_images)
cv2_imshow(grid)
```

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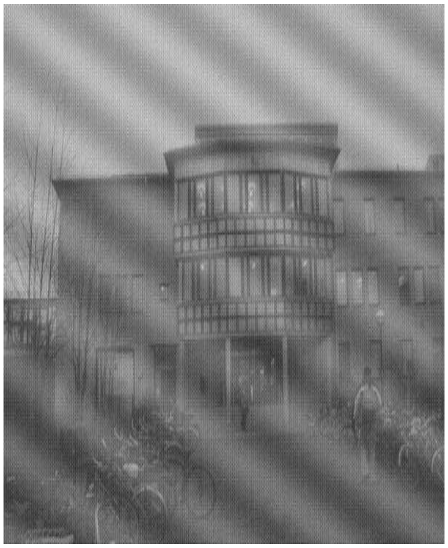
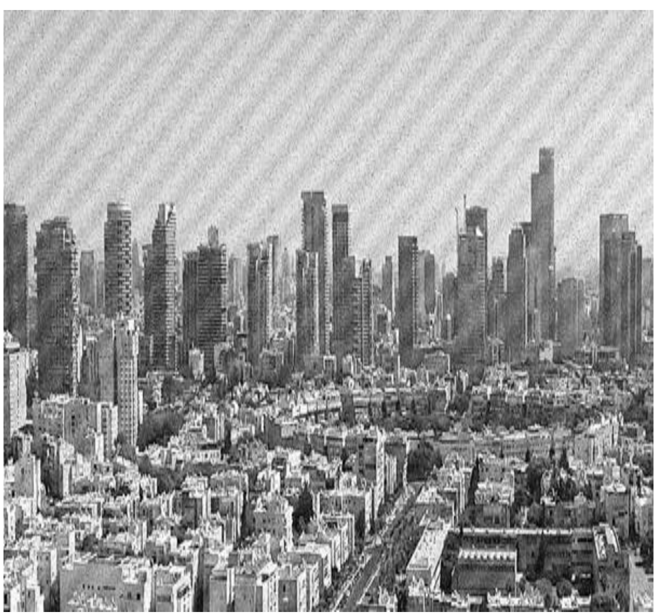
<u>Original Image:</u>





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Output image: